

Organizing Data & Measures of Center

Sections 2.3, 1.7, and 3.1

Stem-and-Leaf Displays

Summation Notation

Mean, Median, Mode, and More

Textbook

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2.3 Stem-and-Leaf Displays

What is a stem-and-leaf display?

A stem-and-leaf display presents **quantitative data in condensed form**.

- ▶ Each observation is divided into a **stem** and a **leaf**.
- ▶ Leaves for each stem are shown separately.
- ▶ Individual observations are not lost, unlike grouped frequency tables.
- ▶ The display helps us see how the data are distributed.

Key idea

A stem-and-leaf display summarizes the distribution while preserving the original observations.

Source: P. S. Mann, *Introductory Statistics*, 10th ed., Wiley, 2020.

Example 2-8: Statistics Test Scores

The scores of 30 college students are:

75	52	80	96	65	79	71	87	93	95
69	72	81	61	76	86	79	68	50	92
83	84	77	64	71	87	72	92	57	98

Question

How can we organize these scores so that we can see their distribution and still keep every score?

We will build the display step by step.

Constructing the Display

For a two-digit number:

stem = first digit	leaf = second digit
--------------------	---------------------

Examples:

$$52 \rightarrow 5 \mid 2 \quad 75 \rightarrow 7 \mid 5$$

Reading the display

$$7 \mid 5 = 75.$$

The scores range from 50 to 98, so the stems are 5, 6, 7, 8, and 9.

Ranked Stem-and-Leaf Display

5		0	2	7					
6		1	4	5	8	9			
7		1	1	2	2	5	6	7	9 9
8		0	1	3	4	6	7	7	
9		2	2	3	5	6	8		

- ▶ Leaves are arranged in increasing order.
- ▶ Stem 7 has the highest frequency.
- ▶ Every original score can be reconstructed.

Source: P. S. Mann, *Introductory Statistics*, 10th ed., Wiley, 2020.

Reading a Stem-and-Leaf Display

For example,

$$8 \mid 0 \quad 1 \quad 3 \quad 4 \quad 6 \quad 7 \quad 7$$

represents

$$80, 81, 83, 84, 86, 87, 87.$$

What can we learn?

- ▶ Number of observations in a range
- ▶ Where observations are concentrated
- ▶ Smallest and largest observations
- ▶ Individual observations, not just frequencies

A stem-and-leaf display is both a summary and a record of the data.

Grouped Stem-and-Leaf Displays

Sometimes there are too many stems, with very few leaves per stem.

Idea

Group consecutive stems together to make a more condensed display.

For example:

0–2

3–5

6–8

An asterisk separates the leaves belonging to different original stems within a grouped row.

Purpose

Grouping can make the overall distribution easier to see when there are many stems.

Split Stem-and-Leaf Displays

If there are only a few stems but many leaves per stem, we can split each stem.

Two-part split

3		0–4
3		5–9
4		0–4
4		5–9

For five parts, the leaf groups are 0; 1–2; 3–4; 5–6; 7–9.

Important

An empty split-stem row indicates that there are no observations in that interval.

Try It: Stem-and-Leaf Display

Use the following data:

41, 28, 45, 60, 53, 69, 70, 50, 63, 68

37, 44, 42, 38, 74, 53, 66, 65, 52, 64

26, 45, 66, 35, 43, 44, 39, 55, 64, 54

- 1 Identify the stems.
- 2 Construct the stem-and-leaf display.
- 3 Arrange the leaves in increasing order.

Work with your group for a few minutes.

1.7 Summation Notation

Suppose a sample contains $x_1, x_2, \dots, x_n$.

$$\sum x = x_1 + x_2 + \cdots + x_n$$

Example:

175, 80, 165, 97, 88

$$\sum x = 175 + 80 + 165 + 97 + 88 = 605.$$

Read it as

“the sum of all values of x ”

Do Not Confuse These Two Expressions

$$\boxed{(\sum x)^2} \quad \text{versus} \quad \boxed{\sum x^2}$$

For $x = 2, 3, 4$:

$$(\sum x)^2 = (2 + 3 + 4)^2 = 81$$

while

$$\sum x^2 = 2^2 + 3^2 + 4^2 = 29.$$

Key distinction

$(\sum x)^2$: add first, then square.

$\sum x^2$: square each value, then add.

If the data are stored in $L1$, the calculator can verify summation notation.

$$\text{sum}(L1) \longleftrightarrow \sum x$$

$$(\text{sum}(L1))^2 \longleftrightarrow (\sum x)^2$$

Squaring the list first gives $L1^2$, and then:

$$\text{sum}(L1^2) \longleftrightarrow \sum x^2.$$

Quick check

Enter a short list in $L1$ and verify that $(\sum x)^2$ and $\sum x^2$ are different.

3.1 Measures of Center for Ungrouped Data

A **measure of center** is a numerical summary describing the center of a data set.

mean, median, trimmed mean, weighted mean, mode

Ungrouped data

The individual values of the data set are available rather than being summarized in a frequency distribution.

Today: understand what each measure tells us and how outliers matter.

3.1.1 Mean

The mean, or arithmetic mean, is obtained by dividing the sum of all values by the number of values.

$$\bar{x} = \frac{\sum x}{n}$$

$$\mu = \frac{\sum x}{N}$$

- ▶ $\bar{x}$ = sample mean
- ▶ μ = population mean
- ▶ n = sample size
- ▶ N = population size

Interpretation

The mean uses every observation in the data set.

Example 3-2: Population Mean

A small company has eight employees with ages:

53, 32, 61, 27, 39, 44, 49, 57.

These are all eight employees, so this is a population:

$$N = 8.$$

$$\sum x = 362.$$

Therefore,

$$\mu = \frac{362}{8} = \boxed{45.25 \text{ years}}.$$

Example 3-1: Sample Mean

For 10 companies, the sum of the 2014 profits is

$$\sum x = 160,703 \text{ million dollars.}$$

Because the 10 companies form a sample,

$$n = 10.$$

Thus,

$$\bar{x} = \frac{160,703}{10} = \boxed{16,070.3 \text{ million dollars}}.$$

Notation

Population: μ Sample: $\bar{x}$

Outliers and the Mean

An **outlier** is an extremely large or extremely small value relative to the rest of the data. For the eight home prices in Example 3-3, the most expensive home is

\$3,874,480,

which is an outlier.

Mean without outlier = \$315,536.29

Mean with outlier = \$760,404.25

Conclusion

The mean can be strongly affected by an outlier. It is a **nonresistant** measure.

3.1.2 Median

The median is the value that divides a ranked data set into two equal halves.

- 1 Rank the data in increasing order.
- 2 Find the value that divides the ranked data into two equal parts.

n odd: median = middle value

n even: median = average of the two middle values

Median: Odd Number of Values

Ranked data:

16.2, 16.9, 19.3, 19.3, 19.6, 21.0, 22.2, 22.5, 28.7, 33.7, 42.1

There are 11 observations, so the single middle value is:

$$\text{Median} = 21.0$$

There are five observations below it and five above it.

Rule

For an odd number of observations, use the one middle value after ranking.

Median: Even Number of Values

Ranked data:

160, 184, 201, 230, 263, 326, 380, 397, 510, 721, 2053, 3864

The two middle values are 326 and 380.

$$\text{Median} = \frac{326 + 380}{2} = \boxed{353}$$

Notice

The two large observations, 2053 and 3864, do not affect the median in the same way they affect the mean.

Mean vs. Median: Outliers

	Mean	Median
Uses all values	Yes	Yes, through ranking
Affected by outliers	Strongly	Much less
Resistant?	No	Yes

Practical idea

When a data set contains substantial outliers, the median may provide a more representative measure of center.

3.1.3 Mode

The **mode** is the value that occurs with the highest frequency.

77, 82, 74, 81, 79, 84, 74, 78

Because 74 occurs twice and every other value occurs once,

Mode = 74 mph

A data set may have no mode, one mode, two modes, or more than two modes.

No Mode, Bimodal, Multimodal

- ▶ If every value occurs once, there is **no mode**.
- ▶ Two values tied for the highest frequency — **bimodal**.
- ▶ More than two values tied for the highest frequency — **multimodal**.

Example:

23, 36, 14, 23, 47, 32, 8, 14, 26, 31, 18, 28

Both 14 and 23 occur twice:

Modes = 14 and 23

Mode for Qualitative Data

The mode can be used for **qualitative as well as quantitative data**.

Example: favorite transportation method:

bus, subway, subway, car, subway, bus, subway

The most frequent category is:

subway

Remember

A numerical average does not make sense for categories, but the most frequent category does.

3.1.4 Trimmed Mean

A $k\%$ trimmed mean is obtained by:

- 1 Ranking the data in increasing order.
- 2 Dropping $k\%$ of the observations from each end.
- 3 Finding the mean of the remaining observations.

For 10 observations and a 10% trimmed mean:

$$10\% \text{ of } 10 = 1.$$

So remove one observation from each end.

Purpose

A trimmed mean reduces the influence of extreme observations.

Example 3-11: 10% Trimmed Mean

Ranked data:

87, 890, 938, 1354, 1429, 1644, 1861, 1993, 2176, 5403

Drop 87 and 5403. The remaining eight values have

$$\sum x = 12,285.$$

Therefore,

$$10\% \text{ trimmed mean} = \frac{12,285}{8} = \boxed{1535.63}.$$

3.1.5 Weighted Mean

When different values have different weights, use:

$$\text{Weighted Mean} = \frac{\sum xw}{\sum w}$$

where x is the value and w is its weight.

Why weights?

A value that represents more observations or a larger quantity should contribute more to the average.

Example 3-12: Gas Prices

Price x	Gallons w	xw
2.60	10	26.00
2.80	13	36.40
2.70	8	21.60
2.75	15	41.25
$\sum w = 46$		$\sum xw = 125.25$

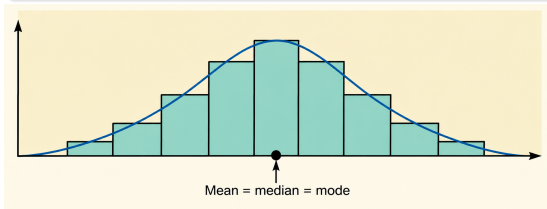
$$\text{Weighted Mean} = \frac{125.25}{46} = \boxed{2.72 \text{ dollars per gallon}}$$

3.1.6 Relationships Among Mean, Median, and Mode

Symmetric distribution

For a symmetric, single-peaked distribution,

$$\text{mean} = \text{median} = \text{mode}$$



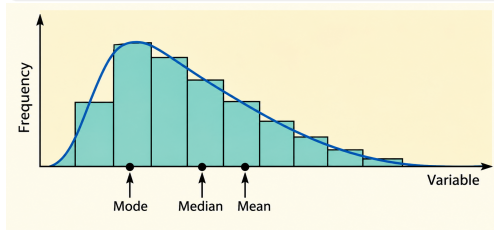
The histogram is balanced on both sides of the center.

Right-Skewed Distribution

Right skew

A long tail to the right pulls the mean toward larger values.

$$\text{mode} < \text{median} < \text{mean}$$



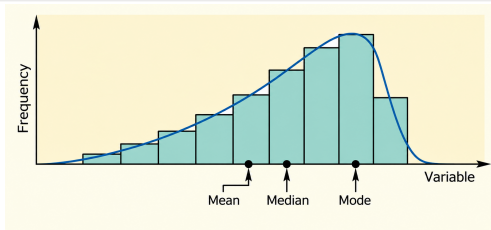
The long right tail pulls the mean to the right.

Left-Skewed Distribution

Left skew

A long tail to the left pulls the mean toward smaller values.

$$\text{mean} < \text{median} < \text{mode}$$



The long left tail pulls the mean to the left.

Which Measure of Center?

Measure	Main idea	Effect of outliers
Mean	Arithmetic average	Strongly affected
Median	Middle of ranked data	Resistant
Mode	Most frequent value	Based on frequency
Trimmed mean	Mean after removing extremes	Less affected
Weighted mean	Mean using weights	Depends on weights

Statistical thinking

No single measure is always best. Consider the data and its distribution.

Quick Check

Choose a measure of center and explain your choice.

- ① A data set contains one extremely large outlier.
- ② A list contains students' favorite colors.
- ③ A data set has no repeated values.
- ④ Different quantities were purchased at different prices.

Think like a statistician

Mean? Median? Mode? Trimmed mean? Weighted mean?

Takeaways

- ▶ Construct and interpret a stem-and-leaf display.
- ▶ Understand grouped and split stems.
- ▶ Use Σ to represent a sum.
- ▶ Distinguish $(\sum x)^2$ from $\sum x^2$.
- ▶ Calculate and interpret the mean, median, and mode.
- ▶ Explain the effect of outliers.
- ▶ Recognize trimmed and weighted means.
- ▶ Relate mean, median, and mode to the shape of a distribution.

Questions?

TI-84: Creating and Editing Lists

Enter data into a list

- 1 Press **STAT**.
- 2 Select **1:Edit**.
- 3 Enter the data values in **L1**.
- 4 Press **ENTER** after each entry.

Example

Enter

18, 7, 46, 32, 62, 65, 51, 40, 28, 72

into **L1**.

Edit a value

Highlight the value, type the replacement, and press **ENTER**.

TI-84: Calculate a Sum

Find $\sum x$

On the home screen:

2nd $\rightarrow$ STAT $\rightarrow$ MATH $\rightarrow$ 5:sum(

Then enter sum(L1) and press **ENTER**.

Example

For $L1 = \{53, 32, 61, 27, 39, 44, 49, 57\}$,

$$\text{sum}(L1) = 362.$$

Connection

$$\text{sum}(L1) = \sum x$$

TI-84: Sums of Squares

Square of a sum

First find **sum(L1)**, then square it:

$$(\text{sum}(\text{L1}))^2 = (\sum x)^2$$

Sum of squares

Create a second list: $L2 = L1^2$ Then calculate: $\text{sum}(\text{L2}) = \sum x^2$

Do not confuse these two quantities

$$(\sum x)^2 \neq \sum x^2$$

TI-84 Practice

Enter in L1

18, 7, 46, 32, 62, 65, 51, 40, 28, 72

- 1 Find **sum(L1)**.
- 2 Find **(sum(L1))²**.
- 3 Create **L2 = L1²**.
- 4 Find **sum(L2)**.

Check

Are $(\sum x)^2$ and $\sum x^2$ equal?

Goal: Be comfortable entering, editing, and working with lists.